

Project ID :

25-26J-467

- Topic (12 words max)

AI-Driven Speech Therapy App

- Research group the project belongs to

SST - Software Systems & Technologies

- Specialization of the project belongs to

Software Engineering (SE)

- If a continuation of a previous project:

Project ID	
Year	

- Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Many children face challenges in speaking clearly due to speech sound disorders, such as difficulties with pronouncing certain sounds or following speech rules. These issues can affect communication, learning, and confidence. According to the American Speech-Language-Hearing Association (ASHA), articulation and phonological disorders are among the most common types of speech difficulties in children (ASHA, n.d.; ASHA, 2023).

Speech therapy is the standard treatment, but access to qualified speech-language pathologists (SLPs) is limited in many regions due to cost, travel distance, or availability. This results in many children not getting enough support or consistent practice. Moreover, traditional methods may not always be engaging for younger users.

To address this issue, our research aims to develop a mobile-based speech therapy app designed specifically for children. The app will use artificial intelligence (AI) and automatic speech recognition (ASR) to listen to a child's speech, detect mispronunciations, and provide instant, helpful feedback. It will include fun and interactive therapy exercises in the form of games to keep children motivated.

From a technical perspective, we aim to train AI models capable of understanding child speech, which varies in tone, speed, and clarity. This poses challenges for standard speech recognition systems, as noted by Ghorbel and Mahdi (2021). Additionally, a study by Vaezipour et al. (2020) highlights the need for speech therapy apps that are more engaging, accurate, and personalized.

Our solution also includes a platform for parents and guardians to monitor their child's progress through reports and summaries. By combining AI with a child-friendly design and caregiver support tools, our research aims to make speech therapy more accessible, effective, and enjoyable for children everywhere.

References

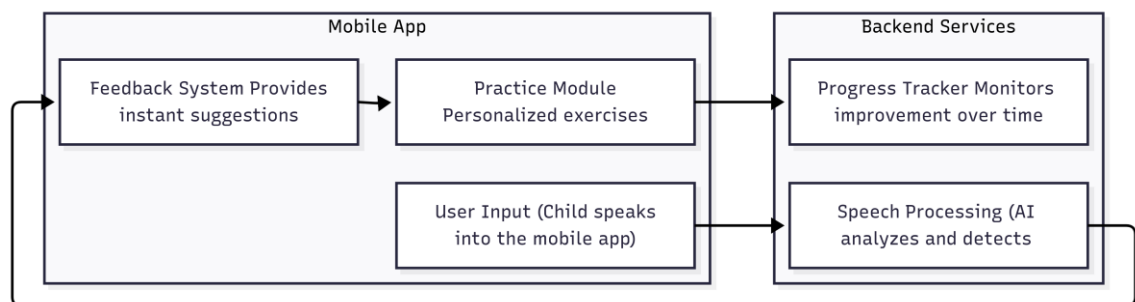
- American Speech-Language-Hearing Association. (n.d.). *Speech Sound Disorders: Articulation and Phonology*. ASHA.
- ASHA. (2023). *Speech Sound Disorders: Articulation and Phonology*. American Speech-Language-Hearing Association.
- Ghorbel, M., & Mahdi, W. (2021). *A survey of automatic speech recognition with deep learning: Trends and challenges*. *Journal of King Saud University - Computer and Information Sciences*.
- Vaezipour A., et al. (2020). *Mobile Apps for Speech-Language Therapy in Adults With Communication Disorders: Review of Content and Quality*. *JMIR Mhealth Uhealth*, 8(10), e18858.

- Brief description of the nature of the solution including a conceptual diagram (250 words max)

We propose a mobile AI-based speech therapy app designed for children with articulation and phonological disorders. Using automatic speech recognition and deep learning, the app analyzes speech to detect common pronunciation errors by comparing it with trained phoneme models.

It provides real-time feedback in a supportive, child-friendly manner and delivers personalized, gamified exercises that adapt based on the child's progress. These interactive activities keep children engaged while practicing difficult sounds. A built-in progress tracker monitors improvement, while a secure dashboard allows parents or caregivers to view reports and receive AI-generated insights to support learning at home.

This solution complements traditional therapy by enabling frequent, engaging practice. By combining intelligent speech analysis, adaptive learning, and child-focused design, the app makes speech improvement more accessible, enjoyable, and effective.



- Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Developing an AI-driven speech therapy app for children requires interdisciplinary expertise combining speech-language pathology, machine learning, and software engineering. While the core team focuses on software development and AI model training, collaboration with speech-language pathologists (SLPs) is essential to understand childhood speech sound disorders, common articulation errors in early development, and age-appropriate therapy practices. This clinical input guides the design of accurate error detection algorithms and effective, child-focused exercises.

From an AI perspective, expertise in automatic speech recognition (ASR), phoneme classification, and deep learning is critical for building models that can accurately process children's speech, which often varies in pitch, clarity, and pronunciation. Knowledge of noise-robust speech processing and real-time feedback mechanisms also contributes to improved system performance and user experience.

The project requires a comprehensive dataset of annotated child speech samples with transcriptions and labeled articulation errors. Publicly available corpora, such as child speech datasets, supplemented by ethically collected, disorder-specific recordings, will be used for model training and evaluation. Metadata on age, dialect, and error types will further enhance the model's ability to generalize across young users.

In addition, robust data privacy and security measures must be implemented to protect sensitive information, particularly as the app is designed for minors and must comply with relevant child data protection regulations.

Combining expertise in pediatric speech pathology, AI technologies, and user-centric app design—along with high-quality, ethically sourced data—is essential to delivering a safe, reliable, and effective AI speech therapy solution for children.

- Objectives and Novelty

Main Objective To design, develop, and evaluate an AI-powered mobile application specifically for children aged 4–12 years that accurately detects speech sound disorders and provides personalized, adaptive therapy through an engaging, gamified interface. The solution will also enable caregivers and clinicians to remotely monitor progress and support each child, thereby improving speech outcomes and accessibility for children with speech disorders.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22356604 - M.H.M Ahamed	To detect phonological and speech sound errors in children, achieving a high accuracy using Trained an AI model.	- Collect and preprocess annotated child speech datasets - Design and train deep learning models for phoneme recognition and general speech error detection - Optimize model for real-time feedback	Age-aware AI model tailored for children's speech, enhancing detection accuracy of phonological errors
IT22900104 - M.Z.M Aathif	To develop an adaptive and personalized therapy module that increases user engagement with customizing exercises, real-time error detection and user progress.	- Collect and preprocess annotated child speech datasets - Design algorithms to tailor therapy exercises dynamically based on detected errors - Implement difficulty adjustment mechanisms responding to user improvement	Real-time adaptive therapy evolving according to each child's learning pace, maximizing motivation and effectiveness

		Collect and analyze user feedback to refine personalization	
IT22609212 - H. Muhammed	-To design a sound based gamified and engaging environment that can boost daily user interaction time, suitable for children with speech disorders. -To design and develop a guardian platform with high reliability, enabling caregivers and therapists to monitor child progress and provide feedback.	- Develop speech-based interactive games with visual and auditory feedback - Implement reward and progress tracking systems - Conduct usability testing with children and refine UI/UX design - Build a secure, intuitive dashboard displaying detailed error data and progress reports	AI-driven gamification personalizes game difficulty and content, sustaining high engagement and practice adherence
IT22235824 - A. N. Pinnawala	To detect articulation errors such as substitution, Omission, Distortions, Additions and give recommendations to correct the speech errors relatively.	- Collect and preprocess annotated child speech datasets - Integrate articulation detection capabilities directly into the app - Ensure compliance with data privacy regulations	Unique integration of articulation detection with a caregiver platform, enabling collaborative, remote support focused on improvements

- Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
IT22356604 - M.H.M Ahamed	AI Model Training for Speech Error Detection: Develop and optimize an AI model to detect phonological and speech sound errors in children aged 4–12, achieving high accuracy and real-time performance.
IT22900104 - M.Z.M Aathif	Adaptive and Personalized Therapy Module: Create a therapy module that dynamically customizes exercises based on real-time speech error detection and user progress to increase engagement and effectiveness.
IT22609212 - H. Muhammed	Gamified, Engaging User Interface & Guardian Platform: Design and implement a child-friendly gamified interface that boosts daily interaction by incorporating interactive speech-based games with multisensory feedback, and develop a secure platform for caregivers and therapists to monitor progress and provide feedback remotely.
IT22235824 - A. N. Pinnawala	Develop and Integrate Articulation Detection: Develop and Integrate articulation detection within the app and give recommendations according to the type of error (substitution, Omission, Additions) that is detected and train them.

- Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Kalpani	Manathunga	
Co-Supervisor	Ms.	Poojani	Gunathilake	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	
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- b) Does the proposed topic exhibit novelty?

Yes		No	
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

[illegible]

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.